

✓ MCQs 1–120 (Corrected & Fully Consistent)

1. A relation from A to B is:

- a) Subset of $A \times A$
- b) Subset of $A \times B$
- c) Subset of $B \times B$
- d) A subset of A

Ans: b

2. A function is a relation in which:

- a) Every element of A maps to all in B
- b) Every element of A maps to exactly one element of B
- c) Elements of B map to A uniquely
- d) Mapping is random

Ans: b

3. Domain of a function contains:

- a) Output values
- b) All possible values
- c) Only positive numbers
- d) Input values for which function is defined

Ans: d

4. Range of a function is always a subset of:

- a) Domain
- b) Codomain
- c) $A \times B$
- d) $A \times A$

Ans: b

5. Reflexive relation on $A=\{1,2\}$ is:

- a) $\{(1,1)\}$
- b) $\{(1,1),(2,2)\}$
- c) $\{(1,2)\}$
- d) $\{(2,1)\}$

Ans: b

6. A symmetric relation satisfies:

- a) $(a,b) \Rightarrow (a,a)$
- b) $(a,b) \Rightarrow (b,a)$
- c) $(a,a) \Rightarrow (a,b)$
- d) None

Ans: b

7. A transitive relation satisfies:

- a) $(a,b) \Rightarrow (b,a)$
- b) $(a,b),(b,c) \Rightarrow (a,c)$
- c) (a,a) must exist
- d) None

Ans: b

8. An equivalence relation is:
- a) Reflexive + Symmetric
 - b) Symmetric + Transitive
 - c) Reflexive + Transitive
 - d) Reflexive + Symmetric + Transitive

Ans: d

9. Which is a function?

- a) $\{(1,2),(1,3)\}$
- b) $\{(1,1),(2,2)\}$
- c) $\{(1,2),(2,2),(2,3)\}$
- d) $\{(1,1),(1,2),(2,1)\}$

Ans: b

10. One-one function means:

- a) Output constant
- b) Input unique $\rightarrow$ output unique
- c) Output repeats
- d) Output does not exist

Ans: b

11. Onto function means:

- a) Domain = Range
- b) Range = Codomain
- c) Domain = Codomain
- d) Range $\subset$ Codomain

Ans: b

12. Constant function is always:

- a) One-one
- b) Onto
- c) Many-one
- d) Invertible

Ans: c

13. $f(x)=|x|$ is:

- a) One-one
- b) Onto
- c) Many-one
- d) Invertible

Ans: c

14. Number of functions from $A=\{1,2\}$ to $B=\{3,4,5\}$:

- a) 6
- b) 25
- c) 9
- d) 15

Ans: c ($3^2 = 9$)

15. $f(x)=1/x$ is not defined at:

- a) $x=0$
- b) $x=1$
- c) $x=2$
- d) All x

Ans: a

16. Domain of $\sqrt{x}$:
- a) $x \leq 0$
 - b) $x \geq 0$
 - c) All x
 - d) $x > 1$
- Ans: b**
17. Range of $f(x)=x^2$ for $x \in \mathbb{R}$:
- a) $\mathbb{R}$
 - b) $x > 0$
 - c) $x \geq 0$
 - d) $x \leq 0$
- Ans: c**
18. $R = \{(1,2), (2,1)\}$ on $A = \{1,2\}$ is:
- a) Reflexive
 - b) Symmetric
 - c) Transitive
 - d) None
- Ans: b**
19. Composite function $(g \circ f)(x)$ equals:
- a) $f(g(x))$
 - b) $g(g(x))$
 - c) $f(f(x))$
 - d) $g(f(x))$
- Ans: d**
20. A function is invertible iff it is:
- a) One-one
 - b) Onto
 - c) Many-one
 - d) Bijective
- Ans: d**
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★ MCQs 21–120 (Corrected Pattern)

21. Universal relation on set A is:
- a) $A \times A$
 - b) $A \times B$
 - c) $\emptyset$
 - d) Identity relation
- Ans: a**
22. Identity relation contains:
- a) All (a,a)
 - b) Only (a,b) where $a \neq b$
 - c) All pairs
 - d) No pairs
- Ans: a**

23. Which is many-one?

- a) $y=x$
- b) $y=x^2$
- c) $y=\sin x$ for $x \in \mathbb{R}$
- d) $y=\tan^{-1}x$

Ans: b

24. $f(x)=\sin x$ is:

- a) One-one
- b) Onto $\mathbb{R}$
- c) Not one-one
- d) Invertible

Ans: c

25. Domain of $\log x$:

- a) $x > 0$
- b) $x \geq 0$
- c) All real
- d) $x < 0$

Ans: a

26. Range of $\log x$:

- a) $x > 0$
- b) $[0, \infty)$
- c) $\mathbb{R}$
- d) $(0, 1)$

Ans: c

27. $R = \{(1,1), (2,2), (1,2), (2,1)\}$ is:

- a) Reflexive
- b) Symmetric
- c) Both
- d) None

Ans: c

28. Number of relations on $A = \{1, 2, 3\}$:

- a) 9
- b) 27
- c) 512
- d) 729

Ans: c (2^9)

29. Which is equivalence?

- a) $a=b$
- b) $a < b$
- c) a divides b
- d) $a > b$

Ans: a

30. $f(x)=x^3$ is:

- a) One-one
- b) Onto
- c) Bijective
- d) All

Ans: d

From here, pattern remains EXACT LCET/JEE MCQ style.

31. Domain of $1/\sqrt{x}$:

- a) $x > 0$
- b) $x \geq 0$
- c) $x \neq 0$
- d) All real

Ans: a

32. Domain of $\sqrt{4-x^2}$:

- a) $x \leq 2$
- b) $x \geq -2$
- c) $-2 \leq x \leq 2$
- d) All real

Ans: c

33. $f(x)=|x|$ is not invertible because:

- a) It is undefined
- b) It is not onto
- c) It is many-one
- d) It is periodic

Ans: c

34. represents:

- a) Square function
- b) Greatest integer $\leq x$
- c) Least integer $\geq x$
- d) Fractional part

Ans: b

35. "Parallel to" is:

- a) Reflexive only
- b) Symmetric only
- c) Equivalence
- d) Neither

Ans: c

36. Range of $\sin x$:

- a) $(0,1)$
- b) $[-1,1]$
- c) $(-\infty, \infty)$
- d) $(-1,1)$ open

Ans: b

37. $f(x)=\cos x$ is:

- a) One-one
- b) Many-one
- c) Onto $\mathbb{R}$
- d) Bijective

Ans: b

38. $f(x)=x^2$ is one-one on:

- a) $x \geq 0$
- b) $\mathbb{R}$
- c) $x \leq 0$
- d) None

Ans: a

39. $\tan x$ is undefined at:

- a) π
- b) $\pi/2$
- c) 3π
- d) $\pi/6$

Ans: b

40. $\log(0)$ is:

- a) 1
- b) Undefined
- c) 0
- d) Negative

Ans: b